

# A primer on Measurement System Analysis

UGAIN – Black Belt in Six Sigma

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<sup>1</sup> P.L. Ottoy – UGent – EA18: ISyE

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Additional information is available on the notes pages.

This document is largely based on :

- [1] “Measurement Systems Analysis” reference manual, Automotive Industry Action Group (AIAG), 1994.
- [2] D.J. WHEELER & R.W. LYDAY, “Evaluating the Measurement Process”, SPC Press Inc., Knoxville, 1989.
- [3] W.J. YOUTEN, “Experimentation and Measurement”, NIST Special Publication 672, reprinted 1997.  
(Freely available from NIST.)

More advanced topics like what to do in case of destructive measurements or when the measurement process changes the part, are not treated in this introduction.

Prerequisites for this document : SPC and ANOVA.

# What is a measurement ?

- (1963 C. Eisenhart) A measurement is the assignment of numbers to material things to represent the relations among them with respect to particular properties.
- A measurement is always expressed as a multiple of some unit quantity and therefore always consists of two parts : a number and a measurement unit. E.g. the length of a rod is **2.1 m**.
- A measurement always comes down to a comparison of a quantity one needs to know with a quantity one assumes to know. The latter is called a **standard** or **reference value**. Standards are defined and maintained by institutes for metrology.
- To conduct a measurement, one needs a measuring instrument (gauge) and a measurement method. Very often a measurement requires manual action of an operator. The collection of gauges, methods, personnel, software, environment, ... to obtain measurements is called the **measurement system**.
- Go/NoGo-tests nor simple counting are not considered as measurements.

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Some quantities can be measured directly, e.g., the length of a rod can be obtained by aligning the rod with a ruler. The markings on the ruler represent (fractions of) the length standard. In some situations a direct comparison to the standard is not possible, in the sense that the measuring uses one or more intermediate quantities. Determining the mass of an object can be achieved via direct comparison with a standard mass, using a balance. But when a load cell is being used, the measurement relies on the intermediate quantities electrical resistance and voltage. In the same way, measuring a temperature with a classical thermometer, uses the thermal expansion of some fluid (mercury, alcohol, toluene, ...) in a capillary tube. When using a thermocouple, the measurement is based on the voltage between two metal junctions. But irrespective of whether the measurement is direct or not, the principle remains the same: the measured value is the result of the comparison of the quantity to the standard.

Instrument and method can range from very simple (e.g. using a ruler to determine the length of something) to very sophisticated (e.g. a complex spectrographic procedure to determine the chemical composition of a substance).

# The importance of measuring

- Building of the pyramids, those giant wonders of beauty and accuracy (Egypt 3000 BC) : death penalty faced those who forgot or neglected their duty to calibrate the standard unit of length at each full moon.
- Important progress on metrology was made due to the efforts of scientists. This eventually led to the Industrial Revolution, marking the era of mass production through interchangeable parts.
- Nowadays (\*) in Europe we measure and weigh at a cost equivalent to 2-7% of our combined GNP! In most modern industries the costs bound up in taking measurements constitute 10-15% of production costs.
- Most of us take for granted the existence of the measurement units we use ; their names form an indispensable part of our vocabulary. Measurement data are used more often and in more ways than ever before in all aspects of human activity.

\* Figures late 20<sup>th</sup> century. In the meantime we can expect this to have increased further.

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Measurements are needed anytime when we want to compare material things with respect to some physical characteristic (e.g., length, volume, mass, ...) and the comparison is not possible in a direct way, such as direct physical alignment of the objects to verify – and adjust if needed – that the desired relationship (e.g., both objects have the same size) holds.

The construction of the great pyramids provides one of the earliest examples of a construction that would not have been possible without a well established measuring process. In order to achieve perfect alignment of the stones, each of the teams of stonemasons, had to carve the stones to very precise dimensions. As these teams worked independently, they had to rely on a common standard of length.

Metrology (the art and science of measurements) has existed in some form or another since antiquity. The earliest forms of metrology were simply arbitrary standards set up by regional or local authorities, often based on practical measures such as the length of an arm. The first royal cubit (Dutch: el) was defined as the length of the forearm from elbow to tip of the extended middle finger of the ruling Pharaoh, plus the breadth of his hand. The original measurement was transferred to and carved in black granite. The workers at the building sites were given copies in granite or wood and it was the responsibility of the architects to maintain them.

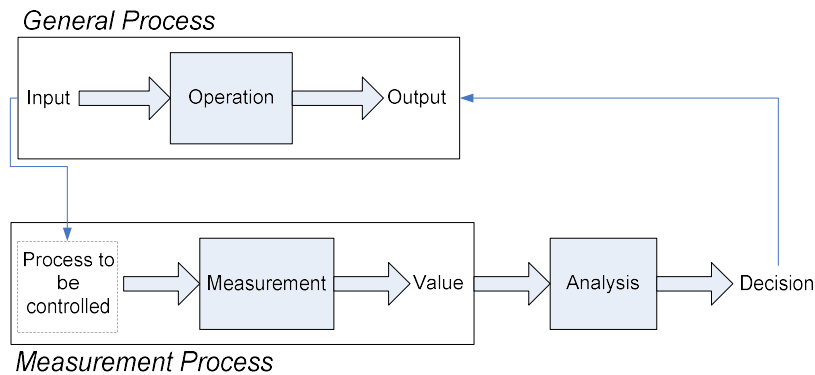
Little progress was made with regard to metrology until various scientists, chemists, and physicists started making headway during the Scientific Revolution (roughly dated from the 16<sup>th</sup> until the 19<sup>th</sup> century). With the advances in the sciences, the comparison of experiment to theory required a rational system of units, and something more closely resembling modern metrology began to come into being. Metrology can be considered as an important precursor to the Industrial Revolution, as it was necessary for the implementation of mass production through the idea of interchangeable parts.

Nowadays the importance of measurements for all aspects of life (trade and commerce, health, authority, leisure time, etc.) can hardly be over-estimated. Just some examples : The pilot carefully observes his altitude, course, fuel consumption and speed, the food inspectorate measures bacteria content, maritime authorities measure buoyancy, companies purchase raw materials by weights and measures, and specify their products using the same units. Processes are regulated and alarms are set off because of measurements. Systematic measurement with known degrees of uncertainty is one of the foundations in industrial quality control. Etc.

We find it almost impossible to describe anything without referring to weights and measures : hours of sunshine, chest measurements, alcohol percentages, weights of letters, room temperatures, tyre pressures ... and so on.

Just for fun, try holding a conversation without using words that refer to weights or measures.

# The measurement process/1



A measurement process may be viewed as a manufacturing process that produces numbers for its output.

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Industry has traditionally viewed the measurement process as a black box. Equipment was the major focus, the more “important” the characteristic, the more expensive the gauge. The usefulness of the gauge, its compatibility with the production process and its environment, was rarely questioned.

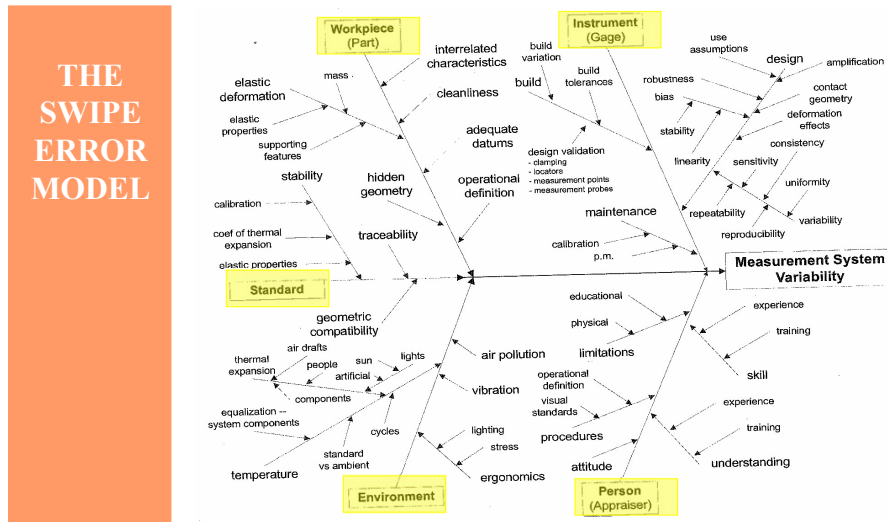
The measuring activity is a process, one producing measurements (numbers), and all of the management, statistical and logical techniques of process control can be applied to it. In industry the customer of the measurement process is the owner of the process to be controlled. This customer’s needs are to be determined first. In general the customer wants to make correct decisions about the process to be controlled with minimum effort (cost). Measuring equipment must be bought which is necessary and sufficient to do this. But equipment is only one aspect. Similar to all processes, the measurement system is impacted by several common and special causes of variation. All these causes have their effect on the performance of the measurement system, making that the measurements vary.

An ideal measurement system would produce only correct measurements. Each measurement would always agree with the reference value. Such a system would be said to have the statistical properties of zero variance, zero bias and zero probability of misclassifying any product it measured (i.e. an ideal gauge performance curve, see further). In general repeated measurements made on the same thing do not agree perfectly. In fact the more sensitive the apparatus, the more surely the variation among repeat measurements will be revealed. Only if a very coarse unit of measurement is used, does the disagreement disappear. E.g. if one would report height only to the nearest whole metre, practically all adult men would be two meters tall. (See further “discrimination”.)

The quality of a measurement system is usually determined solely by the statistical properties of the data it produces over time. Other properties, such as cost, ease of use, etc. are also important in that they contribute to the overall desirability of the system. But it is the statistical properties of the data produced that determines the quality of the measurement system. Therefore, management has the responsibility for identifying the desired statistical properties for the ultimate use of the data. And that these properties are used as the basis for selecting the equipment.

*(continued on the next page)*

# The measurement process/2



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Although each measurement system may be required to have different statistical properties, there are some fundamental properties that belong to any “good” measurement system :

- **Adequate discrimination** : the commonly known Rule of Tens states that instrument discrimination should divide the tolerance or process variation into ten parts or more. This rule is intended as a practical starting point for gauge selection.
- **Statistical stability** : this means that under repeatable conditions, the variation of the measurement system is due to common causes only. Stability can best be evaluated by graphical methods.
- **Variability of the measurement system is small compared to either feature tolerance or process variation** : the observed product or process variability using the measurement system is partly due to the measurement system itself. In a GRR (Gauge Repeatability and Reproducibility) study the measurement system variability is assessed to either feature tolerance or process variation.

The long-standing tradition of reporting measurement error only as a percent of tolerance is inadequate for the challenges of modern industry, where emphasis is on continuous process improvement. As processes change and improve, a measurement system must be re-evaluated for its intended use. It is essential for the organisation to understand the purpose of measurement and apply the appropriate evaluation.

The statistical properties are determined by the sources of variation acting upon the measurement system. In order to control the variation, one needs to identify the sources of variation and control them (or eliminate them if possible).

Although each situation has its specific causes, some typical sources can be identified. The acronym SWIPE represents the six essential elements of a generalized measurement system. SWIPE stands for Standard, Workpiece, Instrument, Person (Procedure) and Environment. Factors affecting these six areas need to be understood so they can be controlled or eliminated. This slide shows a cause-and-effect diagram representation of the SWIPE model. It can be used as a starting point for developing a SWIPE model for a specific measurement system. Working on the factors will lead to measurement data of sufficient quality.

# Measurement terminology : standard

- General definition (NPL)
  - Material measure, measuring instrument, reference material or measurement system intended to define, realise, conserve or reproduce a unit or one or more values of a quantity to serve as a reference.
- Examples :

| Subject field | Standards                                             |
|---------------|-------------------------------------------------------|
| MASS          | mass standards<br>standard balances                   |
| ELECTRICITY   | standard capacitors<br>reference high voltage sources |
| LENGTH        | interferometric comparators<br>gauge blocks           |
| TIME          | cesium atomic clock                                   |

- A standard can be an artifact (e.g. gauge blocks) or an ensemble. An ensemble is the whole of instruments and procedures used in the reproduction of values (e.g. in the calibration of gauge blocks : the comparator, the operator, the environment and the calibration procedure).

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The traceability of measurements is built entirely on standards and calibration procedures.

The notion of standard may have different specific interpretations, depending on the circumstances. E.g. in order of decreasing metrological quality :

A reference standard is a standard of the highest metrological quality available at a given location, from which all measurements made at that location are derived.

A calibration standard serves as a reference in the performance of routine calibrations. In a calibration, strictly following specified conditions, the relation is established between the value indicated by a gauge or the value represented by a material measure and the value realised by a standard. The result of a calibration is written down in a calibration report.

A working standard is used for routine measurements within the laboratory or checks on the shop floor.

Calibration procedures are indispensable to ensure that measuring devices work correctly or that material measures provide correct values. Therefore, a standard at any level has to be compared to a standard at a higher level.

The importance of the notion of standard has already been illustrated with the example of the pyramids. Another striking example, however not a success story in this case, is provided by the famous 17<sup>th</sup> century Swedish warship Vasa. The ship sank on its maiden trip in the harbour of Stockholm after travelling only about one km. The ship was salvaged in the 20<sup>th</sup> century and has since been on display in a museum specially established for that purpose. Although the main reason for the sinking of the ship was the fact that the ship was too high and narrow, with a too high center of gravity, archeologists also uncovered an asymmetry between the port and starboard sides, reducing the stability further. This asymmetry was caused by the use of two slightly different length standards. Regional builders used the Swedish foot (Rikfot  $\approx$  29,69 cm), while also Dutch builders were involved, using the Amsterdam foot ( $\approx$  28,31 cm). This slight difference in standards, made an already bad design even worse. (Source: *Humble Pi: A Comedy of Maths Errors*, Matt Parker.)

# The 7 SI base units

| Unit          | Quantity            | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| s (second)    | Time                | Radiation produced by the transition between the two hyperfine ground states of caesium-133 (in the absence of external influences such as the Earth's magnetic field) has a frequency, $\Delta\nu_{\text{Cs}}$ , of exactly 9 192 631 770 Hz. Hence $1 \text{ s} = 9\,192\,631\,770 / \Delta\nu_{\text{Cs}}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| m (meter)     | Length              | The meter is the length of the path travelled by light in vacuum during a time interval of $1/299\,792\,458$ of a second. The speed of light in vacuum is denoted by $c$ and has the exact value of 299 792 458 m/s. Hence $1 \text{ m} = (9\,192\,631\,770 / 299\,792\,458) \times (c / \Delta\nu_{\text{Cs}})$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| kg (kilogram) | Mass                | The definition of the unit of mass, relies on Planck's constant $h$ , of which the exact value is $6,62607015 \times 10^{-34} \text{ Js} = 6,62607015 \times 10^{-34} \text{ kg.m}^2/\text{s}$ . Hence, using the above definitions for the second and the meter: $1 \text{ kg} = 1,4755214 \times 10^{40} h \times \Delta\nu_{\text{Cs}} / c^2$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| A (Ampère)    | Electric current    | For the definition of the unit of electric current we rely on the value of the electrical charge $e$ of the electron, being $1,602179634 \times 10^{-19} \text{ C}$ . Hence $1 \text{ A} = 1 \text{ C/s} = (10^9 / 1,4728219827) \times e \times \Delta\nu_{\text{Cs}}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| K (Kelvin)    | Temperature         | The unit of temperature depends on Boltzmann's constant $k_B$ being equal to $1,380649 \times 10^{-23} \text{ J/K}$ . From this we derive the definition for the Kelvin: $1 \text{ K} = 2,2666652646 \times h \times \Delta\nu_{\text{Cs}} / k_B$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| mol (mole)    | Amount of substance | Avogadro's constant $N_A$ has the exact value of $6,02214076 \times 10^{23}$ , meaning that one mole of substance consists of exactly this amount of elementary entities (atoms or molecules). Hence $1 \text{ mol} = 6,02214076 \times 10^{23}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| cd (candela)  | Luminous intensity  | The luminous intensity (lichtsterkte) is the luminous flux (lichtflux), unit lumen (lm) through one steradian(*). Hence $1 \text{ cd} = 1 \text{ lm/sr}$ . The energy of photons solely depends on their wavelength or frequency ( $E = h \cdot \nu = h \cdot c / \lambda$ ). However not every light frequency is equally visible for the human eye. The eye is most sensitive for green light. Hence monochromatic light with a frequency of 540 THz (or a wavelength of $\sim 555 \text{ nm}$ ) is chosen as the basis for the definition of the lumen and candela. By convention 683 lm corresponds to the perceived luminous flux of a monochromatic light source with a frequency of 540 THz during one second. Hence $1 \text{ cd} = 8,603341 \times 10^1 \times h \times \Delta\nu_{\text{Cs}} / \text{sr}$ . |

\* The steradian (sr) is an extension of the radian (rd) to 3 dimensions. 1 rd is the angle, with its vertex at the center of a unit circle, that cuts out an arc with length 1. In the same way 1 sr cuts out an area of size 1 on the surface of a unit sphere.

All measurement units can be traced back to the 7 SI base units. These base units are in turn defined in terms of fundamental physical constants, being:

- the frequency of some specific form of radiation emitted by the Caesium-133 isotope ( $\Delta\nu_{\text{Cs}} = 9\,192\,631\,770 \text{ Hz}$ ),
- the speed of light in vacuum ( $c = 299\,792\,458 \text{ m/s}$ ),
- Planck's constant ( $h = 6,62607015 \times 10^{-34} \text{ Js}$ ),
- Avogadro's number ( $N_A = 6,02214076 \times 10^{23}$ )
- the charge of the electron ( $e = 1,602179634 \times 10^{-19} \text{ C}$ )
- Boltzmann's constant ( $k_B = 1,380649 \times 10^{-23} \text{ J/K}$ )

The values of these physical constants mentioned between brackets, have been determined through state of the art measurements and are formally accepted as the exact values within the SI.

While the second is the only base unit to be defined explicitly in terms of the caesium element, all the SI base units (with the exception of the mole) have a definition that either mentions the second or some other units using the second. Therefore the current set of definitions for the SI base units is referred to as the caesium standard. This standard was introduced in 1983.

Any measurable physical property that has a unit, can be expressed in terms of these base units. For example:

- velocity  $\rightarrow \text{m/s}$
- force  $\rightarrow \text{kg.m/s}^2$
- energy  $\rightarrow \text{kg.m}^2/\text{s}^2$
- ...

Evidently this doesn't hold for physical quantities that are unitless by definition, such as refractive index, strain, ...

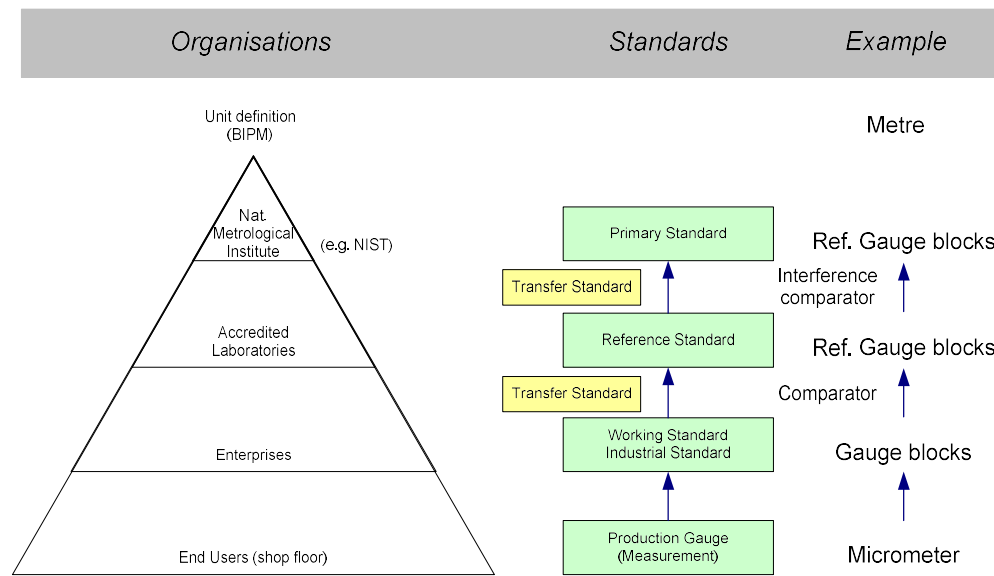
Although reduction is possible, it may not always be useful to do so explicitly, therefore we sometimes use derived units. These derived units improve clarity and practicality. Examples:

- force – derived unit = Newton (N):  $1 \text{ N} = 1 \text{ kg.m/s}^2$
- pressure (force per unit of area) – derived unit = Pascal (Pa):  $1 \text{ Pa} = 1 \text{ kg}/(\text{m.s}^2)$
- energy – derived unit = Joule (J):  $1 \text{ J} = 1 \text{ kg.m}^2/\text{s}^2$
- ...

The current SI system is the result of a long evolution of scientific and technological progress, of which achievements like the Laser Interferometer Gravitational-Wave Observatory (LIGO), the Large Hadron Collider (LHC) and the James Webb telescope are culminations. The system is maintained and further developed by the Bureau des Poids et Mesures (BIPM).

The document “MSA – meter unit.docx” provides additional information on the history of the meter standard.

# The traceability chain



In metrology it all comes down to equality of measurement units. This means that e.g. a meter measured in Belgium should be equal to a meter measured anywhere else in the world. Although this is never exactly true, differences should be acceptably small, not leading to practical problems.

Traceability is defined by the ISO International Vocabulary of Basic and General Terms in Metrology as *The property of a measurement or the value of a standard whereby it can be related to stated references, usually national or international standards, through an unbroken chain of comparisons all having stated uncertainties.*

(The notion of measurement uncertainty will be discussed further. For a fair comparison of measures, the international agreement holds that all indicated measures and tolerances hold for a temperature of 20 °C. The technical aspects of traceability fall outside the scope of this introduction.)

A traceability chain is an unbroken chain of comparisons which shall make it certain that a measurement result or the value of a standard is related to references at a higher level, ending at the final level with a primary standard. In the comparisons each higher level standard serves as a **master** for the subsequent levels.

Primary standards are maintained at National Measurement Institutes (NMI) like NIST (National Institute of Standards and Technology – US NMI belonging to the Department of Commerce). Measurements that can be connected back to NIST through an unbroken chain of comparisons are said to be traceable to NIST. Measurements that are traceable to the same primary standard will agree more closely than those that are not traceable.

Not all organisations have metrology or gauge laboratories within their own facilities and therefore depend on commercial laboratories to provide traceability to an NMI. This is an acceptable practice provided that the capability of the laboratory can be assured through processes like laboratory accreditation (cf. ISO 17025).



# Measurement terminology : reference value

- The true value of an artifact is unknown and unknowable.
- The **reference value** (or master value) is used as a surrogate for the true value.
- As a consequence, both terms are commonly used interchangeably. This is however not recommended.
- Accepted reference values are based upon :
  - Averaging of several measurements with a higher-level measuring equipment.
  - Scientific principles (cf. the SI units of measurement).
  - Legal values : defined and mandated by law.
  - Agreement values : values agreed upon by all affected parties.

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In all cases the reference value should be based on an **operational definition**. “This is one that people can do business with. An operational definition of safe, round, reliable or any other quality [characteristic] must be communicable, with the same meaning to the vendor as to the purchaser, with the same meaning yesterday and today to the production worker” (free quotation from Deming’s famous work “Out of the crisis”).

The measurement system used to determine a reference value should :

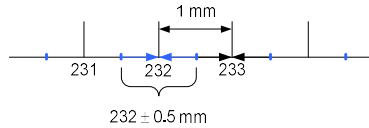
- Include instruments with a higher order discrimination and lower measurement error than the instruments used for normal evaluation.
- Be calibrated with standards traceable to an NMI (like NIST).

Gauge blocks used to calibrate micrometres, have a reference value traceable to an NMI ; i.e. determined with higher level measuring equipments.

As an example of a legally determined reference value : the nautical mile = 1852 m.

# Measurement terminology : discrimination

- **Discrimination** is the amount of change from a reference value that an instrument can detect and faithfully indicate.
- This is typically the value of the smallest graduation on the scale of the instrument : smallest readable unit (**measurement resolution** or **readability**).



- **Effective resolution** : sensitivity of a measurement system to process variation. Sensitivity is defined as the smallest input that results in a detectable output signal. The effective resolution can be larger than the smallest scale graduation (see further “Effect of measurement system error on discrimination”).
- Discrimination is always reported as a unit of measure.

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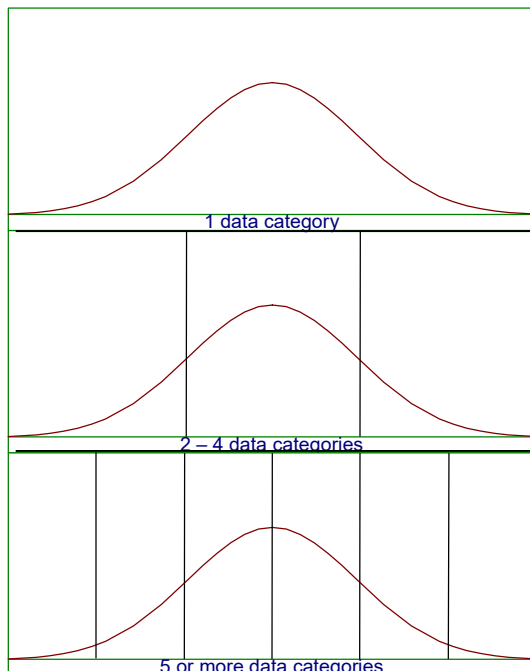
In case of a dial or a ruler, the measuring scale and operating range are obvious. More and more digital measuring equipment is being used, e.g., a kitchen scale with a digital screen. In this case the readability is the least significant digit of the output (e.g., 1/10 of a gram).

In making scientific measurements, it is standard practice to use half-graduations, even to estimate positions in steps of one tenth of the scale interval. Experts may estimate the position on a scale to one twentieth of the scale interval. Very often a small magnifying glass is used as an aid in making these readings.

**A general rule of thumb is that the instrument discrimination ought to be at least one-tenth of the range to be measured. This range is either the product specification or the process variation.**

Factors that affect sensitivity are (among others) skill of appraiser, **repeatability** of the gauge, environment conditions (dirt, humidity, etc). Just because a kitchen scale has a display accurate to 1/10th of a gram does not mean that the actual results vary by 1/10th. A caliper with Vernier scale (or nonius) has typically a readability of 0,02 mm. However, the effective resolution depends in the first place on the skills of the user.

Because of economic or physical limitations, the measurement system will not always perceive all parts of a process distribution as having different values. Instead, the measured characteristic will be grouped by the measured values into data categories.



- Can only be used for process control if process variation is very small compared to specifications ; loss function is flat and main source of variation causes a shift.
- Unacceptable for estimation of process distribution parameters.

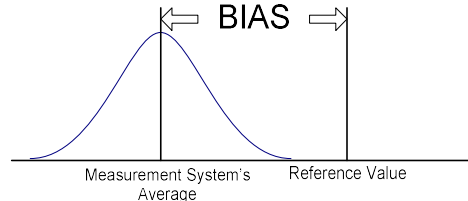
- When used for process control : can lead to false alarms on control charts (see example further).
- Generally unacceptable for estimation of process distribution parameters.

- Fit for variables control charts.
- Recommended for process analysis.

See also “Inadequate discrimination in process control”.

# Measurement terminology : bias

- Bias is the difference between the reference value and the observed average of measurements on the **same** characteristic on the **same** part. It is the systematic error component of the measurement system.
- Potential sources of bias :
  - Gauge needs calibration
  - Error in master
  - Poor gauge design
  - Error in applied constant
  - Fixture error (machining)



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Wheeler[2] extends the notion of bias as follows : A bias effect exists when some identifiable component of measurement error (such as operators or equipment) gives rise to measurements that differ by some fixed amount. Bias effects are named after the particular component of measurement error associated with the bias. That is, when different operators display detectably different average values for measurements of the same thing, an Operator Bias is said to exist. Whenever different equipments display detectably different average values, an Equipment Bias is said to exist.

In a classic GRR study (see further) the reproducibility component is linked to operator bias. The ANOVA method (see further) is appropriate to figure this out.

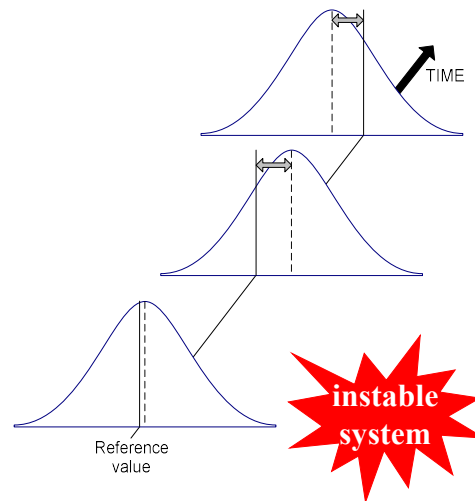
A notion related to bias is accuracy (in some organisations accuracy and bias are used interchangeably). Accuracy is the generic term denoting the closeness of agreement between the measured results and the reference value. ISO and ASTM (American Society for Testing and Materials) use accuracy to denote both bias and repeatability.

ASTM is an international standards organisation (like ISO) founded in 1898 to address one of the major problems of the era : all too frequent rail breaks. ASTM is not a national standards body. In the US this role is taken up by ANSI (American National Standards Institute).

See further "Determining bias".

# Measurement terminology : stability

- Stability describes the change of bias over time. I.e. **a stable measurement system has a constant bias over time.**
- A stable measurement system is in statistical control with respect to location.
- Possible sources of instability :
  - gauge needs calibration
  - wear and aging
  - error in master
  - poor maintenance
  - different method



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Wheeler's notion of consistency encompasses stability (see consistency further). Wheeler's definition of stability is in accordance with the definition of stability from the International Vocabulary of Basic and General Terms in Metrology : "The ability of a measuring instrument to maintain constant its metrological characteristics with time".

See further "Determining stability and consistency".

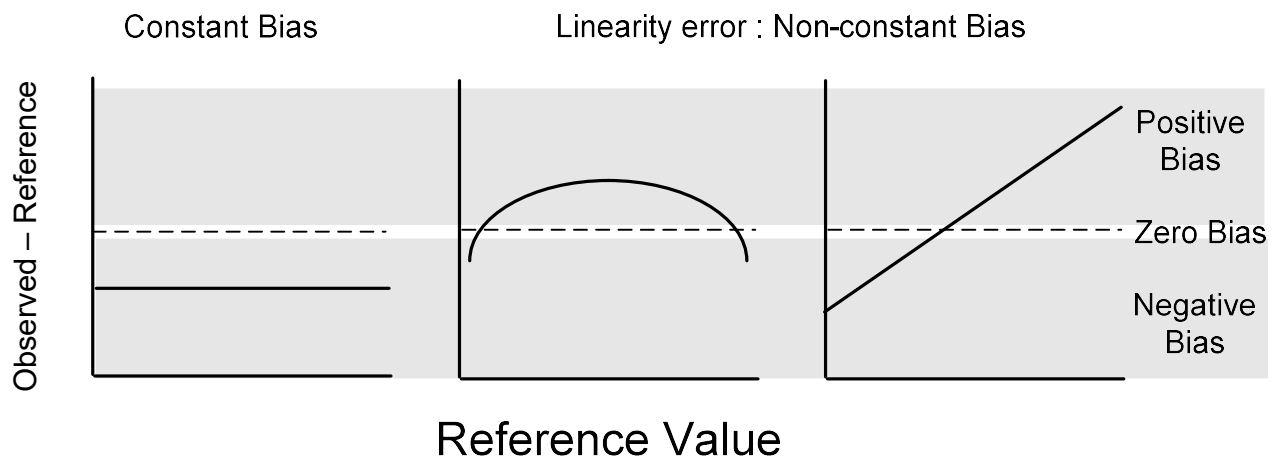
# Measurement terminology : linearity

- Linearity describes the change of **bias** over the **operating range** of the equipment
- Possible sources of linearity error include :
  - gauge needs calibration
  - wear and aging
  - error in master
  - poor maintenance
  - different method (setup, loading or clamping)



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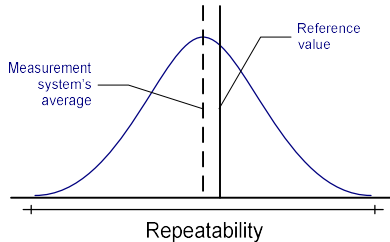
Unacceptable linearity can come in a variety of appearances. The following figures serve as an illustration.



See further "Determining linearity".

# Measurement terminology : repeatability

- Traditional definition : Repeatability is the variation in measurements obtained with **one gauge** when used several times by **one appraiser** while measuring the **identical** characteristic on the **same part** (within appraiser variability).
- Synonyms : Equipment Variation (EV) ; Test-Retest Error.



- Extension : Common cause variation from successive trials under defined conditions of measurement : fixed part, gauge, standard, method, operator, environment, etc. In addition to within-equipment variation, repeatability will include all within variation from any condition in the error model (see higher : SWIPE). **Within-system variation**.
- Potential sources of EV : (within part) form, taper, surface finish ; (within gauge) quality, maintenance, wear ; (within appraiser) skill, experience, technique ; etc.

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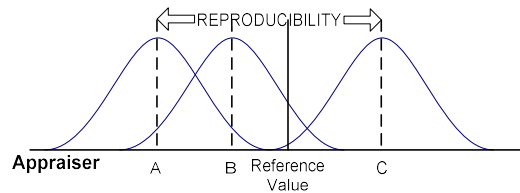
The term **precision** is related to repeatability (in some organisations both are used interchangeably). In fact, precision is a more generic term indicating the “closeness” of repeated readings to each other. ASTM defines precision in a broader sense to include the variation from different readings, gauges, appraisers, labs or conditions.

In the ANOVA-model total variation is decomposed into variation due to underlying factors (explained variation) and a rest term describing the unexplained variation. When this model is applied to measurement variation repeatability stands for the unexplained variation while reproducibility (see next slide) stands for the explained variation. Application of the ANOVA-model to measurement variation is described further.

See further “Determining repeatability and reproducibility”.

# Measurement terminology : reproducibility

- Traditional definition : Reproducibility is the variation in the average of the measurements made by **different appraisers** using the **same** gauge when measuring the **identical** characteristic on the **same** part (between appraisers variability).
- Synonym : Appraiser Variation (AV)
- Potential sources of AV : training, skill, experience, technique, etc.

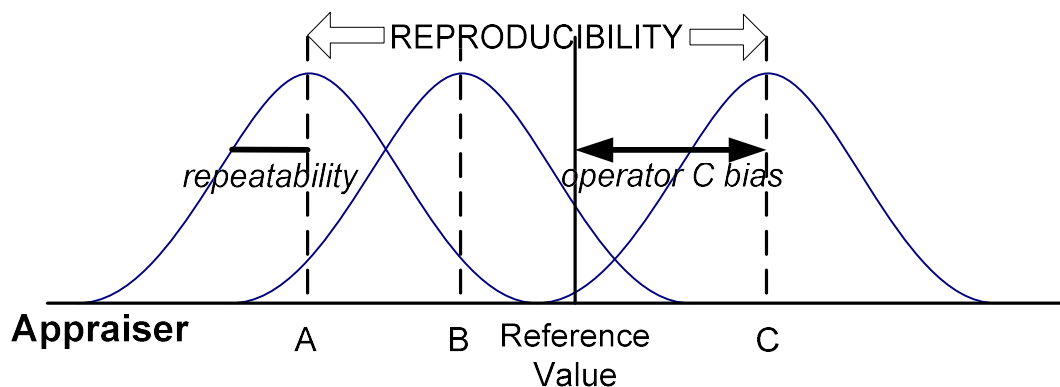


- Extension : Average **variation between systems** or between conditions of measurement (e.g. gauges, labs, environment).
- To have estimates of different components of reproducibility (between operators, between parts, between instruments) one needs measurements following an orthogonal design.

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The extended definition is due to ASTM. Example : With some in-process measurement systems there are no human appraisers. If all the parts are handled, fixtured and measured by the same equipment, then reproducibility is zero, i.e. only repeatability has to be estimated. If however multiple fixtures are used, then the reproducibility is the between-fixture variation.

It is clear that the variation in the average measurements by different appraisers is influenced by the repeatability of the measurement system (\*). A significant reproducibility however is the consequence of the appraiser effect. This implies the existence of Operator Bias (see extended bias definition of Wheeler [2]).



(\*) Even when there would not be any difference between the averages obtained by different appraisers, the repeatability would lead to different (non significant) observed averages.

See further "Determining repeatability and reproducibility".

# Measurement terminology : consistency

- A consistent measurement process has **constant repeatability** over time.
- Consistency, regarded as a measurement system error, is the change in repeatability over time.
- Consistency relates to repeatability as stability relates to bias.
- Wheeler[2] extends this definition to encompass bias, i.e. in this definition stability is part of consistency.
- Sources that may contribute to the consistency error include : temperature of parts, wear of gauge, etc.

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Wheeler[2] defines consistency as follows : “**A measurement process can be said to be consistent if repeated measurements of the same item displays statistical control**”. This means that both the average and the standard deviation of repeated measurements can be considered as constant, i.e. bias and repeatability are constant. Notice that Wheeler’s definition does not require the reference value of the measured item to be known. The bias of the measurement process cannot be estimated. Still it is possible to conclude whether the bias is constant or not.

Wheeler, following Deming, warns not to adjust the measurement process as long as the measurements show statistical control. Adjustment of the measurement process will merely increase variation in the measurements. (Cf. Deming’s funnel experiment, described in “Out of the crisis”.)

See further “Effect of measurement system error on process decisions” and “Determining stability and consistency”.



## Measurement terminology : uniformity

- A uniform measurement process has **constant repeatability over its operating range**.
- Uniformity, regarded as a measurement system error, is the change in repeatability over the operating range of the gauge.
- Uniformity relates to repeatability as linearity relates to bias.
- Sources that may contribute to the uniformity error include: part fixture, parallax reading of scale, etc.

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The notion of uniformity is related to homoscedasticity in linear regression.

## Measurement terminology : gauge R&R (GRR)

- Gauge R&R (GRR) is an estimate of the combined variation of **repeatability** and **reproducibility**.
- $GRR^2$  is the sum of within-system and between-system variance. In formula :

$$\sigma_{GRR}^2 = \sigma_{repeatability}^2 + \sigma_{reproducibility}^2$$

or

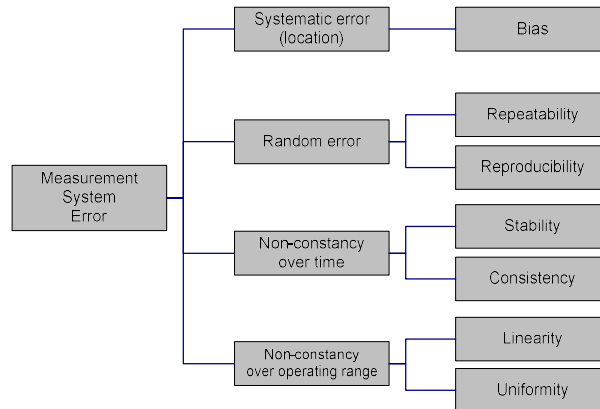
$$GRR = \sqrt{EV^2 + AV^2}$$

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See further “Determining repeatability and reproducibility”.

# Types of measurement system variation

- Classification of measurement system error



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This slide provides a summary of the types of measurement system variation discussed before.

In the industrial practice it is often assumed that measurements are exact. Frequently analysis and conclusions are based upon this assumption as people fail to recognise that variation in the measurement system affects individual measurements and subsequently the derived decisions.

One of the objectives of a measurement system study is to obtain information relative to the amount and types of measurement variation. This information is valuable, since for the average production process, it is **far more practical to recognise the measurement error and establish reasonable limits for it, than to provide extremely accurate gauges**. Applications of such a study are :

- A criterion to accept new measuring equipment.
- A comparison of one measuring equipment against another.
- A basis for evaluating a gauge suspected of being deficient.
- A comparison for measuring equipment before and after repair.
- A required component for calculating process variation, and the acceptability level for a production process.
- Information necessary to develop a Gauge Performance Curve.

# Measurement system capability and performance

- Measurement system **capability** considers measurement system error over a short period of time. It is the combination of errors quantified by linearity, repeatability and reproducibility.

$$\sigma_{cap}^2 = \sigma_{lin}^2 + \sigma_{GRR}^2$$

- Measurement system **performance** is the effect of all sources of variation over time. It adds stability and consistency to the measurement system capability.

$$\sigma_{perf}^2 = \sigma_{cap}^2 + \sigma_{stab}^2 + \sigma_{consist}^2$$

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Estimates of capability and performance are always associated with a defined scope of measurement. The scope for measurement capability e.g. could be a very specific or a general statement of operation, over a limited portion or entire measurement range. Short-term could mean :

- over a series of measurement cycles
- the time to complete a GRR evaluation
- a specific period of production
- time represented by the calibration frequency.

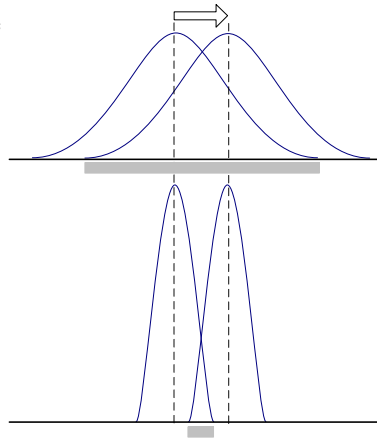
A statement of measurement capability need only be as complete as to reasonably replicate the conditions and range of measurement.

# Effect of measurement system error on discrimination

Effective resolution depends on the measurement variability.

A small change of ref. value is covered up by the measurement error.

Averaging several repeated measurements will confidently demonstrate the change.



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In simple cases the effective resolution is determined by the readability of the gauge. E.g. if a length is measured with a ruler having a scale in mm, a change of the ref. value with 1 mm will always be clearly indicated in the measurement result (assumed that the ruler is cautiously positioned and that the person carrying out the measurement, has no problem in reading the scale).

Measurement system error may lead to an effective resolution that is larger than the discrimination of the gauge. As an illustration, suppose that a particular measurement is made to the nearest 0.1 mm but has a measurement distribution with a standard deviation of 1 mm. It is clear that a change of the ref. value by 0.1 mm will probably go unnoticed in the measurement error. If the measurement error cannot be reduced, the detection of small differences requires several repeated measurements that have to be averaged. This is illustrated on this slide. (How many measurements would be needed here to confidently detect a difference of 0.1 mm ?)

Being able to determine small differences in respect to some property is a major problem in science and industry. This constantly drives the effort towards the development of more accurate measurement systems. Youden[3] (chapter 2) illustrates the need for the determination of small differences with several examples. One striking case concerns the discovery by Lord Rayleigh of the noble gases.

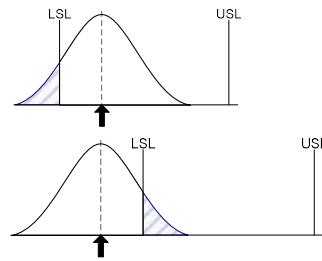
Lord Rayleigh was studying the density of nitrogen gas. He noticed that nitrogen extracted from air was always a tiny bit heavier than nitrogen derived chemically (e.g., from a compound like ammonia). This suggested that something else was mixed with the nitrogen in the air. Rayleigh teamed up with William Ramsay. They took a sample of air and systematically removed every known component (oxygen, carbon dioxide, water vapor and nitrogen). What was left behind was a tiny amount of gas that did not react with anything they tried. It was completely inert. They studied the leftover gas: measured its density, analyzed its spectral lines, ... They concluded it was a new element. They named it argon, from the Greek "argos" (lazy, inactive). (1894) Argon makes up 0,93% of Earth's atmosphere.

# Effects of measurement system error on product decisions/1

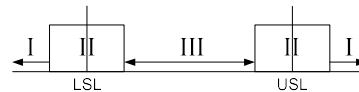
- Wrong decisions are possible whenever the measurement distribution overlaps a specification limit

Type-I error = false negative  
producer's risk

Type-II error = false positive  
consumer's risk



- In general there are three distinct areas
  - I : bad parts will always be called bad
  - II : potential wrong decision can be made
  - III : good parts will always be called good



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We will restrict ourselves to the situation where the measurement serves as a basis for determining the status of a part, with two possibilities :

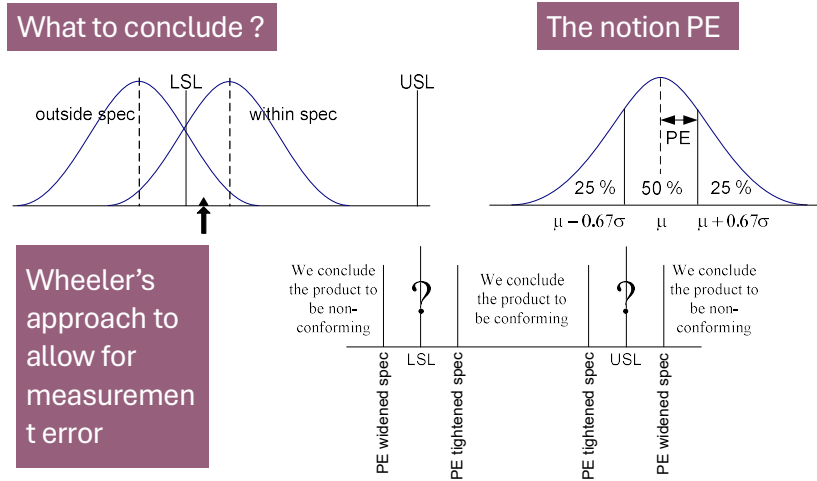
- acceptable = within specification
- unacceptable = outside specification.

To maximize correct decisions regarding product status, there are two choices :

- improve the production process : reduce the variability of the process so that all parts will be produced in the III-area
- improve the measurement system : reduce the measurement system variability so that the II-areas become smaller.

To determine the exact sizes of the three areas, we can calculate the so-called Gauge Performance Curve (see further). This assumes the measurement system distribution to be known.

## Effects of measurement system error on product decisions/2



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The gauge performance curve is based on the reference value of the product being measured, but in practice the decision is based on the measured value. One approach to make allowance for the measurement error is described in Wheeler[2]. The approach uses specification limits that have been adjusted by multiples of the Probable Error (PE). The PE equals about  $0.67\sigma_m$ . PE is the difference between the 75 % quantile and the mean  $\mu$  of the measurement distribution. If we assume this distribution to be approximately normal, then about 50 % of the measurements lie between  $\mu \pm PE$ .

The "PE tightened specs" are obtained by adjusting each of the specification limits towards each other by an amount equal to the PE. Likewise, the "PE widened specs" are obtained by adjusting each of the specification limits away from each other by an amount equal to the PE.

Denote the following events :

- **G**(ood) : the ref. value of the product falls within the interval  $[LSL + \Delta, USL - \Delta]$  for some  $\Delta > 0$ . The product **is** conforming.
- **B**(ad) : the ref. value of the product falls outside the interval  $[LSL - \Delta, USL + \Delta]$ . The product **is** non-conforming.
- **D**(oubt) : the ref. value falls in either of the intervals  $[LSL - \Delta, LSL + \Delta]$  or  $[USL - \Delta, USL + \Delta]$ . The product **is** a borderline case.
- **X** : the measured value falls within the interval  $[LSL + PE, USL - PE]$ . We **conclude** the product to be conforming.
- **Y** : the measured value falls outside the interval  $[LSL - PE, USL + PE]$ . We **conclude** the product to be non-conforming.
- **Z** : the measured value falls in either of the intervals  $[LSL - PE, LSL + PE]$  or  $[USL - PE, USL + PE]$ . We **conclude** the product to be a borderline case.

Now as an exercise, use Bayes' theorem to calculate a lower limit for the odds  $P(G|X)/P(B|X)$  and  $P(B|Y)/P(G|Y)$  as a function of  $\Delta$ . (R proves to be very useful for this job.) Assume that there is no measurement bias, and that the product states **G** and **B** are a priori equally likely. In particular, what are the odds when  $\Delta = PE$  ? Repeat the exercise when the specs are tightened and widened by  $2PE$ . What can be concluded ? (What happens to the odds when going from PE to  $2PE$  ?)

When we cannot draw either one of the conclusions "conforming" or "non-conforming", more measurements of the product are needed. These measurements have to be averaged and the average is to be compared to specs which have been tightened and widened by the PE of the average measurement distribution.

**In practice measurements are often characterized as inadequate when  $\sigma_{GRR}$  is above some percentage of the tolerance interval (commonly 20 % or 30 %). Although having a small  $\sigma_{GRR}$  is desirable, the above shows that even when this is not satisfied, measurements may still have some utility.**

# Effects of measurement system error on process decisions/1

$$\sigma_o^2 = \sigma_a^2 + \sigma_m^2 \Leftrightarrow \frac{1}{C_{Po}^2} = \frac{1}{C_{Pa}^2} + \frac{1}{C_{Pm}^2}$$

o = observed process variation, a = actual process variation, m = measurement variation

assume measurement variation due to repeatability and reproducibility

%GRR based on process variation

$$\%GRR = \frac{6\sigma_m^{(*)}}{6\sigma_o} (\times 100\%)$$

$$\Rightarrow C_{Po} = C_{Pa} \sqrt{1 - \%GRR^2}$$

%GRR based on tolerance range

$$\%GRR = \frac{6\sigma_m^{(*)}}{TOL} (\times 100\%)$$

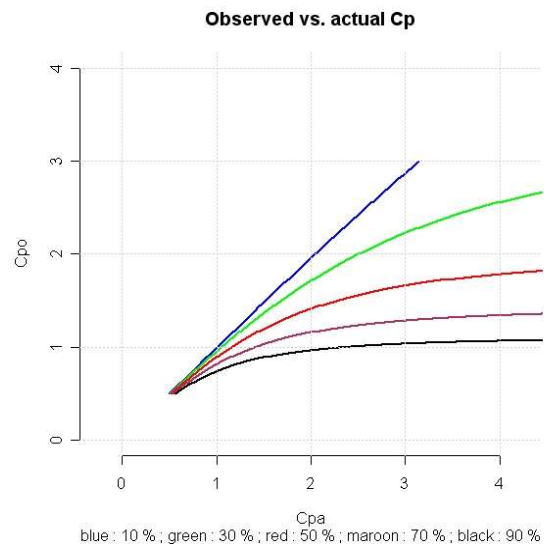
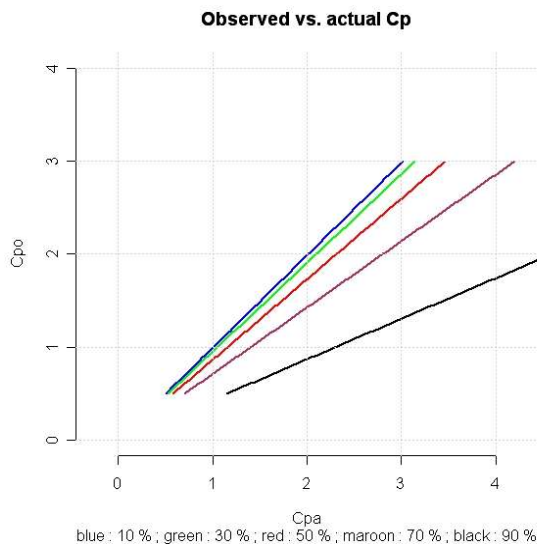
$$\Rightarrow C_{Po} = C_{Pa} \sqrt{1 - (\%GRR \times C_{Po})^2}$$

\* Historically, by convention, a 99 % spread has been used to represent the "full spread of the measurement error". Following this convention, a multiplying factor of 5.15 has to be used. More recently a multiplying factor of 6 is used, representing 99.73 % spread.

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The proof of the formulas is left as an exercise.

The following charts, graphically represent the relation between the actual and the observed  $C_p$  indexes (left : based on process variation ; right : based on tolerance range).



**Exercise :** Suppose a measurement system has a %GRR value of 50 %. What value for the actual process  $C_p$  index would be required in order for the observed  $C_p$  index to be at least 1.33 ? Answer the question for both options to express %GRR.

A  $C_p$  value of at least 1.33 is a common requirement for processes. For a well centered process this implies a defective part rate of not more than 60 ppm.



Proof of formulas on previous slide.

$$C_{p_o} = \frac{USL - LSL}{6\sigma_o} \rightarrow \frac{1}{C_{p_o}} = \frac{6\sigma_o}{USL - LSL} \rightarrow \frac{1}{C_{p_o}^2} = \frac{36}{(USL - LSL)^2} \sigma_o^2 \rightarrow \frac{1}{C_{p_o}^2} = \frac{36}{(USL - LSL)^2} (\sigma_a^2 + \sigma_m^2)$$

$$\rightarrow \frac{1}{C_{p_o}^2} = \frac{1}{C_{p_a}^2} + \frac{1}{C_{p_m}^2}$$

$$\%GRR = \frac{6\sigma_m}{6\sigma_o} = \frac{C_{p_o}}{C_{p_m}} \rightarrow C_{p_o} = C_{p_m} \cdot \%GRR \rightarrow \frac{1}{C_{p_o}^2} = \frac{1}{C_{p_a}^2} + \frac{\%GRR^2}{C_{p_o}^2} \rightarrow C_{p_o}^2 = C_{p_a}^2 (1 - \%GRR^2)$$

## Effects of measurement system error on process decisions/2

- Effect on new process acceptance

When a new process (e.g. machining, stamping, heat treating, assembling, etc) is purchased, this often involves some studies done on the equipment, first at the supplier's location and then at the customer's location. If the measurement systems used at either location is not consistent with that used under normal circumstances, then confusion may ensue.

- Effect on process control

Often manufacturing operations use a single part at the beginning of the day (shift) to verify the process is targeted. If the part measured is off target, then the process is adjusted. This type of measurement and decision making normally adds extra variation to the process. (Deming referred to this practice as tampering.)

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We illustrate both effects with an example.

### Process acceptance

Suppose during buy-off a high order measurement system is used, with a %GRR = 10 % (based on tolerance range). An actual process  $C_p$  of 2 leads to an observed  $C_p$  of 1.96. When this process is studied in production with the production gauge, more variation will be observed. Suppose the %GRR of the production gauge is 30 %. When the actual process  $C_p$  is still 2, the observed process  $C_p$  will be no more than 1.71 (but still acceptable). A worst case scenario would be if a production gauge has not been qualified, but is nevertheless used. Suppose the measurement system GRR is actually 60 % (but this fact is not known), then the observed  $C_p$  would only be 1.20. A value unacceptable in most cases. This difference between the observed  $C_p$  at buy-off and the observed  $C_p$  during production is fully due to the measurement system and not to any form of process deterioration.

### Process control

Consider a situation where the weight of a precious metal coating on a part is being controlled to a target of 5.00 grams. Suppose that the results from the scale used to determine the weight vary  $\pm 0.20$  grams but this is not known since the measurement system analysis was never done. The operating instructions require the operator to verify the weight at setup and further every hour, based on one production part. If the result is beyond the interval 4.90 to 5.10 grams, then the operator is to setup the process again.

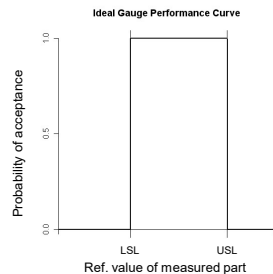
At a given control point, suppose the process is operating at 4.95 grams, but due to measurement error, the operator observes 4.85 grams. According to the instructions, the operator attempts to adjust the process up by 0.15 grams. Now the process is running at 5.10 grams for a target. When the operator checks again, he observes 5.08 grams (due to measurement error) so the process is allowed to run. Over-adjustment of the process has added variation to it and will continue to do so.

Measurement system error forms a natural limit to the (observed) process variation. When the process variation approaches this limit, it becomes hard (or even impossible) to determine the actual process variation. This is due to the fact that the actual process variation can only be found by subtracting the measurement system error from the observed process variation. When both these variances become similar, their difference becomes quite small. The uncertainty of both variances will generally overwhelm the small difference, so that the estimate of the actual process variation will be quite unreliable.

# Gauge performance curve

- A gauge performance curve (GPC) determines the probability of accepting a part of some reference value, i.e.  $GPC : X_r \rightarrow \beta(X_r)$ .
- A GPC can be compared to an OC-curve.

⇒ An ideal GPC looks like this :



⇒ Calculation of the GPC requires knowledge of the measurement system error.

⇒ Example :

$$X_m \sim N(X_r + b, \sigma_{GRR})$$

$$\beta(X_r) = \Phi\left(\frac{USL - (X_r + b)}{\sigma_{GRR}}\right) - \Phi\left(\frac{LSL - (X_r + b)}{\sigma_{GRR}}\right)$$

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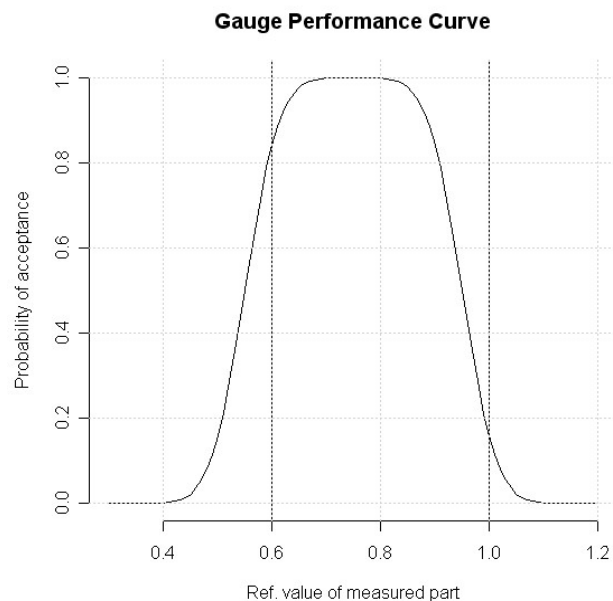
Suppose a given gauge is used to measure some characteristic of parts in a production process. Comparison of the measurement  $X_m$  to the specification limits (LSL and USL) leads to either acceptance or rejection of a part. The GPC characterises the influence of the measurement system error on this decision process (see also “Effects of measurement system error on product decisions”). Without measurement system error, the GPC takes its ideal form, and the probability of committing a decision error is 0. In reality measurement system error is not zero, and decision mistakes will occur.

In the example it is assumed that measurement system error is due to bias ( $b$ ) and lack of repeatability and reproducibility (represented by  $\sigma_{GRR}$ ). Furthermore the shape of the distribution is assumed to be normal. Proof as an exercise that the probability of accepting a part with given reference value  $X_r$  is given by the formula on the slide ( $\Phi$  is the standard normal cumulative distribution function).

The GPC on the right holds for : LSL = 0.6 ; USL = 1.0 ;  $b = 0.05$  and  $\sigma_{GRR} = 0.05$ .

What is the probability of accepting a part with ref. value 0.5 ? (This is a part outside the specification limits.)

What is the probability of accepting a part with ref. value 0.9 ? (This is a part within the specification limits.)



# Measurement uncertainty/1

- Measurement uncertainty describes the quality of a measurement value, in that it identifies the range around the measurement value that, with a defined level of confidence, is expected to contain the reference value. In formula :

$$\text{Ref. value} = \text{measurement} \pm U$$

where :

$U$  = expanded uncertainty =  $k \cdot u_c$

$k$  = coverage factor ; representing the level of confidence

$u_c$  = combined standard uncertainty representing all significant components of variation in the measurement process ;  $u_c = \sigma_{\text{measurement error}}$

- Measurement uncertainty is a **confidence interval for the reference value based on the measurement.**

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Higher we stated that any measurement consists of two parts : a number and a measurement unit. However, where accurate measurement is crucial, a more subtle definition is required to encompass the uncertainty associated to the measure. In this definition there are three parts: first, the measurement itself, second, the margin of error, and third, the confidence level -- that is, the probability that the actual property of the physical object is within the margin of error.

A measurement uncertainty statement has always an associated scope with it, identifying all significant errors taken into account. E.g. if only repeatability and reproducibility have been taken into account, based on a short-term assessment, then  $\sigma_{me} = \sigma_{GRR}$ .

The ISO Guide to the Uncertainty in Measurement specifies that the coverage factor should correspond to a level of confidence equal to 95 %. In practice the value  $k = 2$  is used.

Example :

length = 20 mm  $\pm$  1mm, at a level of confidence of 95 %.

# Measurement uncertainty/2

- To calculate the uncertainty of a measurement, firstly identify the sources of uncertainty, then estimate the size of the uncertainty from each source, finally combine the individual uncertainties to get the combined uncertainty. The following rules apply :
- Uncertainties are estimated by either a type A or a type B evaluation :
  - type A : using statistics
  - type B : using other information like calibration certificates, manufacturer's specifications, calculations, common sense, ...
- Apply the following formulas to combine uncertainties :

|                                          |                                                                                     |
|------------------------------------------|-------------------------------------------------------------------------------------|
| $u_c(\bar{x}) = \frac{u(x)}{\sqrt{n}}$   |                                                                                     |
| $u_c(x+y) = \sqrt{u^2(x) + u^2(y)}$      | $\frac{u_c(x \cdot y)}{x \cdot y} = \sqrt{\frac{u^2(x)}{x^2} + \frac{u^2(y)}{y^2}}$ |
| $\frac{u_c(x^2)}{x^2} = \frac{2u(x)}{x}$ | $\frac{u_c(\sqrt{x})}{\sqrt{x}} = \frac{u(x)}{2x}$                                  |

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The formulas show how measurement uncertainty propagates when applying a function to the measured values. Suppose a quantity  $x$  is measured and the uncertainty is  $u(x)$ . If we use  $x$  in a function  $y = f(x)$ , then the uncertainty will be carried over to  $y$ . It can be shown that (which we accept here without proof):

$$u(y) \approx |f'(x)| \cdot u(x)$$

The derivative  $f'(x)$  tells how sensitive  $y$  is to changes in  $x$ . The more sensitive, the larger the propagation of the uncertainty. As an example suppose  $f(x) = x^2$ . Then

$$u(x^2) \approx 2x \cdot u(x)$$

or equivalently

$$\frac{u(x^2)}{x^2} = \frac{2u(x)}{x}$$

when squaring a measured value, the relative uncertainty doubles.

A numerical example. Suppose  $x = 5 \pm 0,1$ . If  $f(x) = x^2$ , then  $f'(x) = 2x$ . Hence  $u(x^2) \approx 10 \times 0,1 = 1$ . And thus  $x^2 = 25 \pm 1$ .

The other formulas can be derived in a similar way.

See also the separate document "Calculating the uncertainty in the width of a steel strip.doc" for an illustration.

# Inadequate discrimination in process control

- Effect
  - Inadequate discrimination can make an  $\bar{X}/R$ -chart appear to be out-of-control, even when the underlying process is stable.
  - Illustration : Rheostat knob data.
- Insufficient discrimination detection rule of thumb :
  - [Subgroup size = 2] If there are not more than 3 possible values for the range within the control limits, then the measurement units are too large for the purposes of the control chart.
  - [Subgroup size  $\geq 3$ ] Idem, but now up to 4 different values.
- Solution
  - Have a measurement system with better discrimination.
  - Increase variation within subgroups to a detectable level, usually by collecting values for a subgroup over a longer time span (applicable in situations with a stable, highly capable process using a stable, best-practice measurement system).

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A fairly widespread measurement problem is when measurement units are too large for process control. To detect this problem, no standard parts or batches are needed, a simple average and range chart suffices. The example contained in “Rheostat Knob Data.xls” shows the effect on the control-chart of inadequate discrimination. The chart appears to be out of control although the process is stable. When this happens, one cannot safely interpret the control chart as reflecting the behaviour of the process.

In situations of highly capable processes, effective resolution, even of best-practice measurement systems, may be inadequate. In such case further improvement of the measurement system may become impractical. The only solution is then to increase within subgroup variation by defining larger subgroups over a longer time span.

The rationale behind the detection rule is as follows :

- Assume a subgroup size  $n = 2$ . Then  $E(R) \cong 1.1\sigma$ . This means that on average the distance between two measurements is about  $1.1\sigma$ .
- Two measurements can be rounded to the same value, only when their difference is not more than one measurement unit (MU).
- Combining both observations : when  $\sigma$  is smaller than one MU, measurements within a subgroup will begin to be rounded to the same value. This rounding will contaminate the distance between the control limits by deflating the average range.
- The smaller  $\sigma$  is relative to the MU, the greater the contamination will be. The case  $\sigma = MU$  is the borderline condition. The rule of thumb presented on this slide, is based on this borderline condition.
- In the borderline case, the formulas for the control limits of the R-chart, are :  $UCL = D_4 \cdot d_2 \cdot MU$  and  $LCL = D_3 \cdot d_2 \cdot MU$ . For  $n = 2$ , we get  $UCL - LCL \cong 3.69MU$ , i.e. the borderline condition states that there are only 4 possible values between the control limits on the range chart. So less than 4 (not more than 3) possible values between the control limits on the range chart, is indicative for inadequate MUs. In the same way, for  $n = 3$  we get  $UCL - LCL \cong 4.36MU$ , i.e. not more than 4 possible values is indicative for inadequate MUs. It is left as an exercise to verify larger subgroup sizes.

# Determining stability and consistency

- Obtain a reference standard (sample of which the reference value is established relative to a traceable standard).
- Periodically measure this standard several times (it is common to take repeated measurements three to five times , by the same appraiser).
- Each time, calculate the subgroup mean and standard deviation or range and plot the result on an  $\bar{X}/s$ - or an  $\bar{X}/R$ -chart.
- If the measurement process is stable, the data can be used to estimate the bias and the repeatability of the measurement system :

$$\text{bias} = \bar{\bar{X}} - \text{ref.value}$$

$$\text{repeatability} = \frac{\bar{R}}{d_2}$$

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Some comments on this procedure :

- If a reference standard is not available, select a production part that falls in the mid-range of production measurements. Hold it apart and designate it as the master for stability analysis. (In this case it is impossible to estimate the bias. Why ?) Caution should be taken where this production master could experience excessive wear due to repeated use and handling.
- It may be desirable to have master samples for the low end, the high end, and the mid-range of the measurements. Separate control charts are needed for each.
- Sample size and frequency should be based on knowledge of the measurement system. Factors to include may be: how often is it required to re-calibrate or repair the gauge, how frequently is it used, how stressful are the operating conditions, etc. The periodic readings should be taken at different times to represent when the measurement system is actually being used. This will account for warm-up, ambient or other factors that may change during the day.
- When the control chart for location is out of control, a simple re-calibration may re-establish the measurement process in a state of statistical control. However, without knowledge of why the measurement process went out of control, one either risks bad values on production parts or continually has to measure the reference standard in conjunction with each production part.
- When the range chart is out of control, there is definitely a problem with the measurement process and one has to search until the assignable cause is found.
- If it is impossible or impractical to repeatedly measure a reference standard, duplicate measurements on production samples may be used to check the consistency (in the strict meaning) of the measurement process.

# Determining bias

- To determine if a gauge exhibits significant bias, the following procedure can be applied based on the  $\bar{X}/R$ -chart data.
- Compute the bias :  $\text{bias} = \bar{\bar{X}} - \text{ref.value}$
- Compute the repeatability :  $\sigma_r = \bar{R}/d_2$
- It then can be shown that the following statistic follows a t-distribution with n degrees of freedom :

$$t = \frac{\text{bias} - E(\text{bias})}{\sigma_r} \cdot \frac{\sqrt{gm} \cdot d_2^*}{d_2}$$

with g the number of subgroups, m the subgroup-size and  $d_2^*$  a so-called conversion factor, related to  $d_2$ .

- Therefore a  $(1 - \alpha) \times 100$  % confidence interval for the true gauge bias is given by :

$$\text{bias} - \sigma_r \frac{d_2}{d_2^* \sqrt{gm}} t_{1-\alpha/2, v} \leq E(\text{bias}) \leq \text{bias} + \sigma_r \frac{d_2}{d_2^* \sqrt{gm}} t_{1-\alpha/2, v}$$

- The gauge bias is significant if zero does not belong to this interval.

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The fact that the t-statistic follows a t-distribution, is based on the following result by P.B. Patnaik (see A.J. Duncan "Quality control and industrial statistics") :

$$\frac{v \bar{R}^2}{d_2^{*2} \sigma^2} \sim \chi^2(v)$$

Patnaik worked out conversion factors  $d_2^*$  and degrees of freedom v for various values of m (subgroup size) and g (number of rational subgroups). Remark that in this case v is not an integer value (the notion of degrees of freedoms can indeed be extended to non-integer values). The values of  $d_2^*$  and v can be found in an appendix to this course.

Given this result by Patnaik, prove as an exercise that the t-statistic follows a t-distribution and derive the formula for the confidence interval for the true gauge bias.

To estimate  $d_2^*$ , we use the fact that  $E(\chi^2(v)) = v$ . Hence  $d_2^{*2} = \frac{E(R^2)}{\sigma^2}$ .



# Determining linearity

- Conducting the study
  - Select  $g \geq 5$  parts whose measurements, due to process variation, cover the operating range of the gauge.
  - Establish the reference value for each part, relative to a traceable standard. Confirm the operating range of the gauge is encompassed.
  - Have each part measured  $m \geq 10$  times with the gauge by one of the operators who normally use the gauge. Select the parts at random to minimize appraiser “recall” bias in the measurements.
- Graphical analysis of the results
  - Calculate the bias for each measurement  $j$  of each part  $i$  :  $\text{bias}_{i,j} = m_{i,j} - \text{ref. value}_i$  ( $m$  = measurement).
  - Plot the biases vs. the ref. values and fit a line through the plotted points applying simple linear regression.
  - Calculate and plot the  $(1 - \alpha) \times 100$  % confidence bands for  $E(\text{bias})$  as a function of the ref. values.
  - Plot the bias = 0 line and review the graph for indications of lack of linearity.

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When the bias = 0 line entirely lies within the confidence bands for  $E(\text{bias})$  we can accept the gauge to have no bias over its operating range.

It may well be possible that the line bias = 0 is not encompassed in the confidence bands, but that instead a line bias =  $b$  (for some nonzero value  $b$ ) does fall entirely between the confidence bands. In this case the gauge satisfies the criterion of linearity but is not free of bias.

Testing linearity comes down to testing the hypothesis  $H_0$  : slope = 0 vs.  $H_A$  : slope  $\neq 0$  for the regression line. When the null hypothesis can be accepted, the gauge satisfies linearity. (Beware of lack of fit however : always plot the biases vs. the ref. values.) In this case the additional hypothesis test  $H_0$  : intercept = 0 vs.  $H_A$  : intercept  $\neq 0$  provides further information on the possible presence of constant bias over the operating range. When bias is present, its magnitude may be estimated by calculating a  $(1 - \alpha) \times 100$  % confidence interval for the intercept.

## Exercise :

A plant supervisor was introducing a new measurement system to the process. As part of the PPAP (Production Part Approval Process, part of the QS 9000 standard) the linearity of the measurement system needed to be evaluated. Five parts were chosen throughout the operating range of the measurement system based upon documented process variation. Each part was measured by layout inspection to determine its reference value. Each part was then measured twelve times by the lead operator. The parts were selected at random during the study. The results are contained in the file “linearity.txt”. Based on these data investigate whether the gauge satisfies the linearity criterion or not.

# Determining repeatability and reproducibility

## Average and range method/1

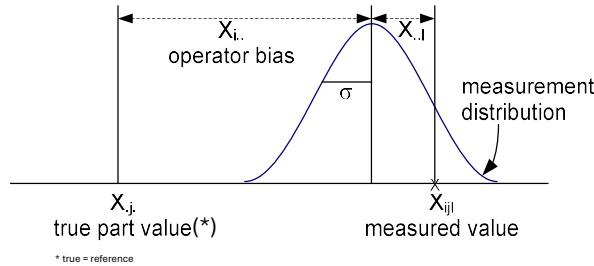
### The model

k appraisers each measure n parts r times with the gauge. The parts are representative for the process variation.

$$X_{ijl} = X_{j.} + X_{i..} + X_{..l}$$

$$X_{..l} \sim N(0, \sigma)$$

$\sigma$  = gauge repeatability



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From this model the following averages and ranges are derived :

Calculation of the averages :

$\bar{X}_i$

average measured value by appraiser i over all parts and replications

$$\begin{aligned}\bar{X}_i &= \frac{1}{nr} \sum_j \sum_l X_{ijl} = \frac{1}{nr} \sum_j \sum_l (X_{j.} + X_{i..} + X_{..l}) = \frac{1}{n} \sum_j X_{j.} + X_{i..} + \frac{1}{nr} \sum_j \sum_l X_{..l} \\ &= \bar{X}_p + X_{i..} + \bar{\bar{X}}_r\end{aligned}$$

$\bar{X}_p$  is a fixed number for the given parts,  $\bar{\bar{X}}_r$  is the average measurement error over all measurements by appraiser i and is linked to the repeatability  $\sigma$  by

$$\bar{\bar{X}}_r \sim N(0, \sigma/\sqrt{nr})$$

$\bar{X}_j$

average measured value for part j over all appraisers and replications

likewise, we obtain :

$$\bar{X}_j = X_{j.} + \bar{X}_a + \bar{\bar{X}}_r$$

$\bar{X}_a$  is a fixed number for the given appraisers,  $\bar{\bar{X}}_r$  here is the average measurement error over all measurements of part j and is linked to the repeatability  $\sigma$  by

$$\bar{\bar{X}}_r \sim N(0, \sigma/\sqrt{kr})$$

Calculation of the ranges :

$\bar{\bar{R}}$

average range over all parts and appraisers

$$\bar{\bar{R}} = \frac{1}{nk} \sum_i \sum_j R_{ij}$$

$\bar{X}_{DIFF}$

range of the appraiser averages

$\bar{R}_p$

range of the part averages

# Determining repeatability and reproducibility

## Average and range method/2

### Estimation of the repeatability EV

$$E(\bar{R}) = d_2 \sigma \Rightarrow EV = \sigma \cong \bar{R}/d_2 = \bar{R} \times K_1$$

$$K_1 = 1/d_2 ; d_2\text{-value based on subgroupsize} = r$$

### Estimation of the reproducibility AV

$$E(\bar{X}_{DIFF}^2) = d_2^{*2} \sigma^2(\bar{X}_i) \Rightarrow \sigma^2(\bar{X}_i) \cong \bar{X}_{DIFF}^2/d_2^{*2} = (\bar{X}_{DIFF} \times K_2)^2$$

$$K_2 = 1/d_2^* ; d_2^*\text{-value based on 1 subgroup of size} = k$$

$$\Rightarrow AV = \sqrt{(\bar{X}_{DIFF} \times K_2)^2 - EV^2/(nr)}$$

### Estimation of the part variation PV

$$\text{likewise } PV = \sqrt{(\bar{R}_p \times K_3)^2 - EV^2/(kr)} \approx \bar{R}_p \times K_3$$

$$K_3 = 1/d_2^* ; \text{based on 1 subgroup of size} = n$$

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The formulas for AV and PV are based on the result by Patnaik, see before.

In the formula for PV : if process variation is large compared to equipment variation and if the selected parts properly represent process variation, then the formula for PV can be simplified to  $\bar{R}_p \times K_3$

AV and EV together determine the gauge repeatability and reproducibility GRR (see slide 18) :  $GRR^2 = EV^2 + AV^2$ . All sources of variation taken together determine the total variation TV :  $TV^2 = GRR^2 + PV^2$ .

An important criterium to determine the adequacy of the gauge is based on the quantity  $\%GRR = GRR/TV \times 100\%$  (see also slide 24). The following rule of thumb is often used to judge on the adequacy of the gauge :

|                            |                                                                                                        |
|----------------------------|--------------------------------------------------------------------------------------------------------|
| $\%GRR \leq 10\%$ :        | gauge generally considered to be acceptable                                                            |
| $10\% < \%GRR \leq 30\%$ : | gauge may be acceptable based upon importance of application, cost of procurement, cost of repair, etc |
| $30\% < \%GRR$ :           | gauge considered to be not acceptable – every effort should be made to improve the measurement system  |

# Determining repeatability and reproducibility Anova

⇒ The data in a GRR study enable the calculation of a two-way ANOVA-table with factors appraiser and part. For a model without interaction, the table looks like :

|                 | Factor        | SS     | df                                  | MS                 | F                 |
|-----------------|---------------|--------|-------------------------------------|--------------------|-------------------|
| reproducibility | a = appraiser | $SS_a$ | $df_a = (k - 1)$                    | $MS_a = SS_a/df_a$ | $F_a = MS_a/MS_e$ |
|                 | p = part      | $SS_p$ | $df_p = (n - 1)$                    | $MS_p = SS_p/df_p$ | $F_p = MS_p/MS_e$ |
| repeatability   | e = error     | $SS_e$ | $df_e = nk(r - 1) + (k - 1)(n - 1)$ | $MS_e = SS_e/df_e$ |                   |
|                 |               | $SS_T$ | $nkr - 1$                           |                    |                   |

⇒ Significance of the factor appraiser means there is an appraiser bias effect, i.e. different appraisers have different bias (reproducibility significantly > 0).

⇒ The quantities EV, AV and PV can be calculated from the table as follows :

$$EV = \sqrt{MS_e}; AV = \sqrt{(MS_a - EV^2)/(nr)}; PV = \sqrt{(MS_p - EV^2)/(kr)}$$

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In the ANOVA approach, the notion of reproducibility refers to the factors identified in the model : firstly this is the factor appraiser (traditional), but by extension this also refers to the factor part. The notion of repeatability refers to the error component in the model : measurement variation that cannot be explained by any of the factors. Therefore the advantages of ANOVA over the traditional average and range approach are :

- the model can be extended to comprise different components of reproducibility : between appraisers, between parts, between instruments, etc. (see slide 14)
- the model can be extended to allow the study of interactions : e.g. does the appraiser bias depend on the part being measured (such a model explicitly requires measurement replications to estimate the repeatability) ?

Furthermore ANOVA offers more accurate estimates for repeatability and reproducibility.

In the two-way ANOVA model (without interaction) the expected measurement value for a given appraiser i and a given part j is given by :

$$\hat{X}_{ij} = \bar{\bar{X}} + (\bar{X}_i - \bar{\bar{X}}) + (\bar{X}_j - \bar{\bar{X}})$$

therefore the residual (actually measured value – model value) is given by :

$$X_{ijl} - \hat{X}_{ij} = X_{ijl} - \bar{X}_i - \bar{X}_j + \bar{\bar{X}}$$

Using this results, one can easily prove that :

$$SS_T = \sum_i \sum_j \sum_l (X_{ijl} - \hat{X}_{ij})^2 + \sum_i \sum_j \sum_l (\bar{X}_i - \bar{\bar{X}})^2 + \sum_i \sum_j \sum_l (\bar{X}_j - \bar{\bar{X}})^2$$

As a consequence (see also "Theoretical background of a GRR study.pdf"):

$$MS_e = \frac{\sum_i \sum_j \sum_l (X_{ijl} - \hat{X}_{ij})^2}{nk(r - 1) + (n - 1)(k - 1)} = EV^2$$

$$MS_a = \frac{nr \sum_i (\bar{X}_i - \bar{\bar{X}})^2}{k - 1} = nr(AV^2 + \frac{EV^2}{nr})$$

Proof of formula for  $SS_T$ .

$$\hat{X}_{ij} = \bar{\bar{X}} + (\bar{X}_i - \bar{\bar{X}}) + (\bar{X}_j - \bar{\bar{X}})$$

therefore

$$X_{ijl} - \bar{\bar{X}} = (X_{ijl} - \hat{X}_{ij}) + (\bar{X}_i - \bar{\bar{X}}) + (\bar{X}_j - \bar{\bar{X}})$$

This leads to

$$\begin{aligned} SS_T &= \sum_i \sum_j \sum_l (X_{ijl} - \bar{\bar{X}})^2 = \\ &= \sum_i \sum_j \sum_l (X_{ijl} - \hat{X}_{ij})^2 + \sum_i \sum_j \sum_l (\bar{X}_i - \bar{\bar{X}})^2 + \sum_i \sum_j \sum_l (\bar{X}_j - \bar{\bar{X}})^2 \\ &\quad + 2 \sum_i \sum_j \sum_l (X_{ijl} - \hat{X}_{ij}) (\bar{X}_i - \bar{\bar{X}}) \quad (1) \\ &\quad + 2 \sum_i \sum_j \sum_l (X_{ijl} - \hat{X}_{ij}) (\bar{X}_j - \bar{\bar{X}}) \quad (2) \\ &\quad + 2 \sum_i \sum_j \sum_l (\bar{X}_i - \bar{\bar{X}}) (\bar{X}_j - \bar{\bar{X}}) \quad (3) \end{aligned}$$

$$\begin{aligned} (1) &= 2 \sum_i \sum_j \sum_l (X_{ijl} - \bar{X}_i - \bar{X}_j + \bar{\bar{X}}) (\bar{X}_i - \bar{\bar{X}}) = 2 \sum_i (\bar{X}_i - \bar{\bar{X}}) [\sum_j \sum_l X_{ijl} - nr\bar{X}_i - r \sum_j \bar{X}_j + nr\bar{\bar{X}}] \\ &= 2 \sum_i (\bar{X}_i - \bar{\bar{X}}) [nr\bar{X}_i - nr\bar{X}_i - nr\bar{\bar{X}} + nr\bar{\bar{X}}] = 0 \end{aligned}$$

$$(2) \text{ (idem)} = 0$$

$$(3) = 2r \sum_i (\bar{X}_i - \bar{\bar{X}}) \sum_j (\bar{X}_j - \bar{\bar{X}}) = 0$$

## Exercise : GRR study

A mechanical part used in the steering mechanism of trucks, is measured using a jig (to fix it) and several measuring gauges. In fact each part selected for measurement, has nine different dimensions recorded. The following data relate to only one of these dimensions. The question is whether the measurement system is acceptable.

Three operators (one in each production shift) routinely use the gauge and therefore were selected to participate in the study. Five parts were picked at random on each of five different days in order to obtain a cross-section of the product variation.

The dimension is measured to the nearest half-thousandth of an inch (0.0005 inch). The measurements are coded to make them easier for use. The gauge is set up so that a reading of 0 corresponds to an actual value of 1.5400 inch. The data were then recorded in units of 0.0001 inch in order to avoid having decimal values. As an example : a coded value of 45 corresponds to an actual value of  $1.5400 + 45 \times 0.0001 = 1.5445$  inch.

Each operator measured each part twice, giving in total 15 subgroups of size 2. The operators are referred to as A, B and C, the parts are referred to as 1, ..., 5. The measurement procedure is as follows, taking into account the fact that the operators are on different shifts : operator A measures the 5 parts in random order. Then a new random order is determined and the operator repeats the measurements later on during his shift. The same is repeated for operators B and C during their shift.

### Remark :

Measurements are blind : i.e. operators are not aware of the fact that an assessment of the measurement system is being conducted. This is done to avoid contamination of the measurements by the well-known Hawthorne effect. (People behaving better as a result of being part of a study.)

As a consequence operators cannot identify the parts being measured. Identification of the parts is kept secret to the operators.

The results of the study are shown in the file "GRR – ANOVA – average and range – 2.xlsx" on the tab sheet "measurements".

Conduct the GRR analysis and determine whether the measurement system is acceptable.

See file "GRR – ANOVA – average and range - 2.xlsx".